

Comparing Absorbency of Paper Towel Brands

Stage 3: Final Experiment
Implementation Analysis

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Introduction and Motivation

Background

Research Question: Do different paper towel brands have significantly different absorption rates after controlling for liquid type?

Objective: Compare mean absorbency across towel brands and quantify the magnitude of those differences using a generalized complete block design.

Motivation and Background:

Paper towels are essential cleaning products in the foodservice industry, where absorbency directly affects cleaning efficiency, waste, and operational cost. Although many brands advertise superior performance, objective comparisons under controlled experimental conditions are limited.

Why the Experiment Is Interesting or Important:

Understanding whether absorbency differences are statistically and practically significant can help restaurant owners make more informed purchasing decisions by balancing product cost against cleaning performance and efficiency.

Experimental Design Summary

Variables of Interest

Treatment Factor: Brands, (Bounty, Sparkle, and Viva) 3 *levels*

Blocking Factor: Liquid Type, (Water, Soda) 2 *levels*

Treatment Structure: One-way treatment structure

Response Variable: Amount of liquid absorbed by a paper towel sheet in grams.

- **Measurement Method:** Absorbency was measured at the point of first dripping.
- **Units:** Grams (g) absorbed.

Experimental and Measurement Units

- **Experimental Units:**

The experimental units in this experiment are time slots. These are assigned at random to paper towel brands, blocked on each liquid type.

- **Measurement Units:**

The measurement unit would be defined as the paper towel in the respective time slot.

Sources of Variation and Control

Treatment Factor (Major Source of Variation): Paper towel brands.

Nuisance Factors: Temperature and humidity, experimental error, sheet size , rate of liquid application.

Minor Sources of Variation:

- Differences between sheets within the same roll
- Scale/measurement error
- Other uncontrolled experimental factors factors

How Variability will be Controlled:

Blocking: A generalized complete block design (GCBD) was used, with liquid type serving as the blocking factor and towel brands randomized within each block.

Fixing Nuisance Factors:

- All trials were conducted under consistent temperature and humidity conditions.
- Paper towel size and liquid volume were standardized across trials.
- Liquid application and post-absorption weighing procedures were kept consistent for all samples.

Experimental Design and Randomization

Design Structure:

Generalized Complete Block Design

Replicates: Each brand will be tested multiple times within each liquid block to maintain a balanced design to improve precision.

Experimental Layout Summary

Liquid Type (Blocks)	Towel Brand (Treatments)			
	Bounty (B)	Viva (V)	Sparkle (S)	Total per Block
Water	6 samples	6 samples	6 samples	18 samples
Soda	6 samples	6 samples	6 samples	18 samples
Total per Brand	12 samples	12 samples	12 samples	36 samples

Randomization Procedure (for each liquid block)

1 Generate random permutation of 1–18 in R

Example permutation

7, 3, 15, 1, 11, 18, 4, 9, 14,
2, 17, 6, 10, 16, 8, 13, 5, 12

2 Assign numbers to brands (6 each)

Bounty (B)

(numbers 1–6)

7, 3, 15, 1, 11, 18

Viva (V)

(numbers 7–12)

4, 9, 14, 2, 17, 6

Sparkle (S)

(numbers 13–18)

10, 16, 8, 13, 5, 12

3 Sort by numerical value to determine testing order

Final randomized order (by number)

1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18

S S B V V S S V B V V B B S B V S B

4 Follow this sequence to test the other 18 samples

Block 1: Water

18 tests (6 per brand)

Time_Slot (order)	Brand
1	S
2	S
3	B
4	V
5	V
6	S
7	S
8	V
9	B
10	V
11	V
12	B
13	B
14	B
15	S
16	B
17	S
18	V

Block 2: Soda

18 tests (6 per brand)

Time_Slot (order)	Brand
1	S
2	S
3	B
4	S
5	B
6	S
7	V
8	V
9	V
10	B
11	V
12	B
13	S
14	V
15	B
16	V
17	S
18	B

Finalized Experiment Implementation

1. Use the same size of paper towels (6 sheets per brand) with each brands.
2. Within each block, run the randomization. Get the test orders.
3. Weigh the **dry** sheet and record: **DryWeight (g)**.
4. Apply liquid at a consistent rate until the first visible drip occurs from the sheet.
5. Weigh the **wet** sheet and record: **WetWeight (g)**.
6. Compute **Absorbed = WetWeight - DryWeight(g)**.
7. Record results for the respective trial.
8. Dispose, and reset materials for the next time slot.



Data Collection and Implementation

Pilot Study

The pilot study gave us the opportunity to not only simulate the final experiment on a smaller scale, but also to help identify and address potential problems in the data collection. It also helped to obtain a variance for the sample size calculation

Issues Encountered in Pilot:

- Paper towels were held by a person which could introduce human error by tilting the paper towel, fatigue etc.
- We used a whole-integer scale which reduced precision in measurement.

Final Experiment Adjustments

- Chopsticks were used to transfer fully absorbed towels to the scale, minimizing additional moisture loss.
- Measurement precision was addressed by replacing an integer scale with a digital scale accurate to 0.01 g, reducing measurement error.
- Reduced human error by using dumbbells to hold the paper towel rather than a person. This also created a more consistent experiment environment.

Contrast of Interest

- We are interested in all pairwise comparisons among the 3 brands.
- Tukey-based adjustments were used to control the strong family wise error rate. This correction is used for the sample power calculator and the simultaneous confidence intervals.
- μ_B, μ_S, μ_V -the mean absorbed by each brand

$$\mu_B - \mu_S, \mu_B - \mu_V, \mu_S - \mu_V$$

$$\alpha = 0.05$$

$$\Delta = 2g$$

Sample Size Calculation

reps_per_block_treatment	samples_per_block	total_sample_size	power
2	6	12	0.2415030
3	9	18	0.4173137
4	12	24	0.5735973
5	15	30	0.7000781
6	18	36	0.7958747
7	21	42	0.8648899

Where we denote the following:

- F_w = F-test statistic for the contrast
- α = Significance level
- v = treatments
- b = number of blocks
- n = total sample size
- Δ = Target effect size
- r = replicates per block
- δ = noncentrality parameter
- σ = standard deviation (squared is variance)

We also establish the following parameters of use:

- $\pi = 0.80$
- $\alpha = 0.05$
- $\sigma^2 = 2.20$ (pilot study)
- $\Delta = 2$
- $df_{error} = N - b - v + 1$

We conclude that the minimum number of sample size needed is of $s=7$ to achieve a power of minimum 80%.

$$\delta = \frac{br\Delta^2}{2\sigma^2}$$

$$P(\text{reject } H_0 | H_a \text{ is true}) = P(F_W > \frac{q_{\alpha, v, N-b-v+1}^2}{2} | F_W \sim F_{1, N-b-v+1}(\delta))$$

Sample Size Calculation

$$\sum c_i \tau_i \in \left(\sum c_i \bar{y}_{.i.} \pm w \sqrt{msE \sum c_i^2 / bs} \right)$$

Where we denote the following:

- $\bar{Y}$ - sample mean for the treatment within a block
- C - coefficients of the contrast
- W - Tukey critical coefficient
- MSE - mean square error (variance)
- b - number of blocks
- s - sample size for each block in each treatment
- q - studentized range critical value
- α - alpha (Type 1 error)
- v - number of treatments

r	total_n	df	q	w	ME	L
2	12	8	4.041036	2.857444	2.9969129	5.9938257
3	18	14	3.701394	2.617280	2.2413051	4.4826102
4	24	20	3.577935	2.529982	1.8762847	3.7525694
5	30	26	3.514171	2.484894	1.6482921	3.2965842
6	36	32	3.475248	2.457372	1.4880124	2.9760248

We desire to have a target confidence interval length of $L \leq 4$, to obtain a half width of $\Delta \leq 2$ while controlling for tukey multiple comparison.

$$w_T = q_{v,df,\alpha} / \sqrt{2}$$

Experiment Data

Absorbed = Wet-weight minus Dry-weight

Time_Slot	Brand	Dry_Weight	Wet_Weight	Absorbed	Liquid_Type
1	S	1.82	20.17	18.35	water
2	S	1.8	18.96	17.16	water
3	B	2.45	31.86	29.41	water
4	V	2.92	33.69	30.77	water
5	V	2.95	30.7	27.75	water
6	S	1.82	20.71	18.89	water
7	S	1.84	21.15	19.31	water
8	V	3	31.88	28.88	water
9	B	2.46	30.14	27.68	water
10	V	2.97	31.47	28.5	water
11	V	2.93	31.77	28.84	water
12	B	2.42	32.48	30.06	water
13	B	2.46	30.44	27.98	water
14	B	2.46	32.22	29.76	water
15	S	1.82	20.97	19.15	water
16	B	2.47	30.37	27.9	water
17	S	1.82	20.15	18.33	water
18	V	2.93	31.08	28.15	water

Time_Slot	Brand	Dry_Weight	Wet_Weight	Absorbed	Liquid_Type
1	S	1.81	19.87	18.06	soda
2	S	1.83	20.47	18.64	soda
3	B	2.46	30.85	28.39	soda
4	S	1.83	22.7	20.87	soda
5	B	2.47	33.28	30.81	soda
6	S	1.85	20.69	18.84	soda
7	V	2.96	31.55	28.59	soda
8	V	3	34.61	31.61	soda
9	V	2.96	32.85	29.89	soda
10	B	2.46	31.08	28.62	soda
11	V	3	33.67	30.67	soda
12	B	2.43	30.49	28.06	soda
13	S	1.84	21.47	19.63	soda
14	V	3	33.06	30.06	soda
15	B	2.46	33.57	31.11	soda
16	V	3	34.2	31.2	soda
17	S	1.81	20.73	18.92	soda
18	B	2.46	34.23	31.77	soda

Statistical Analysis

Data Analysis Methods Used (Overview)

Exploratory Data Analysis

- Summary statistics and boxplots were used to compare absorbency across towel brands and liquid types and interaction plot was examined .

Initial Model

- A Block-Treatment Model with Brand, Liquid Type, and Brand $\times$ Liquid Type interaction was first fitted.
- The interaction term was tested to determine whether brand performance depended on liquid type.

Final Model

- Since the Brand $\times$ Liquid Type interaction was not significant, main effects were tested.
- Brand effect was significant, thus one of the brand statistically is different from the others.

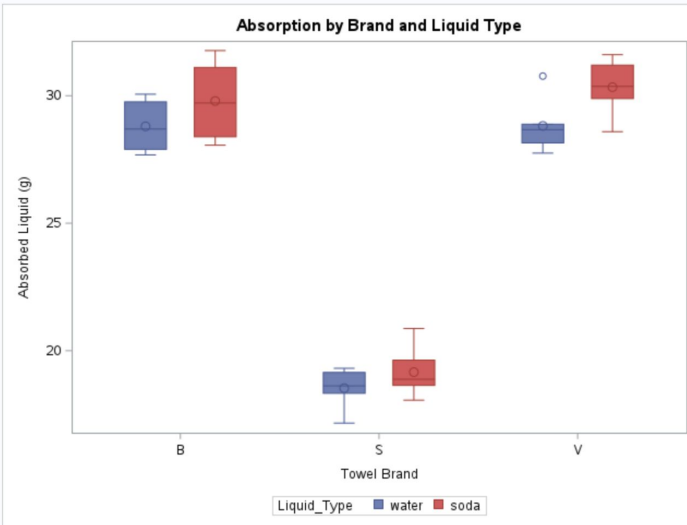
Post-hoc Comparisons

- Tukey-adjusted pairwise comparisons were used to compare all three towel brands while controlling the strong family-wise error rate.

Model Diagnostics

- Residual plots, QQ plot, histogram, and Cook's distance were used to assess model assumptions and influential observations.

Exploratory Data Analysis



The MEANS Procedure

Analysis Variable : Absorbed						
Brand	Liquid_Type	N Obs	Mean	Std Dev	Minimum	Maximum
B	soda	6	29.7933333	1.6139971	28.0600000	31.7700000
	water	6	28.7983333	1.0600079	27.6800000	30.0600000
S	soda	6	19.1600000	0.9783047	18.0600000	20.8700000
	water	6	18.5316667	0.7837963	17.1600000	19.3100000
V	soda	6	30.3366667	1.0774724	28.5900000	31.6100000
	water	6	28.8150000	1.0490520	27.7500000	30.7700000

- Sparkle Brand has the lowest mean absorbency under both liquids.
- Bounty and Viva Brands absorbed substantially more liquid.
- Soda shows slightly higher absorbency than water across all brands

Standard deviations were relatively small within groups (approximately 0.78–1.61 g), indicating low within-group variability and consistent absorption measurements.

Statistical model : General complete block design

$$Y_{hit} = \mu + \theta_h + \tau_i + (\theta\tau)_{hi} + \epsilon_{hit}$$

where

Y_{hit} = response observed on treatment i in block h at replicate t ,

μ = overall mean response,

θ_h = effect of block h ,

τ_i = effect of treatment i ,

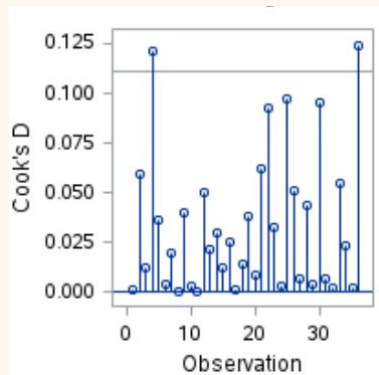
$(\theta\tau)_{hi}$ = block-treatment interaction effect,

ϵ_{hit} = random error term.

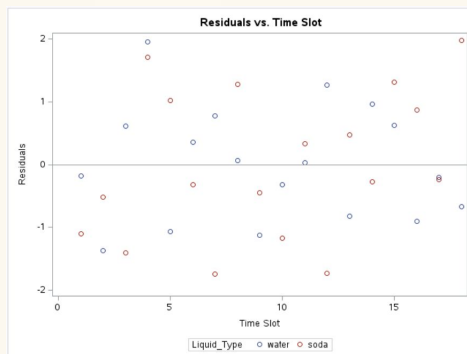
Model Assumptions

- **Outliers / Influence:** Residual plots and Cook's distance were examined for outlier or influential points. Only 2 potential outliers was found.
- **Independence:** Observations were assumed independent due to randomized testing and separate towel measurements.
- **Normality:** Residual errors were assumed approximately normally distributed around zero.
- **Constant Variance:** Residual variability was assumed comparable across brands and liquid types.

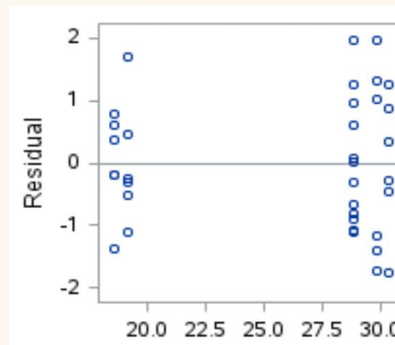
Diagnostics



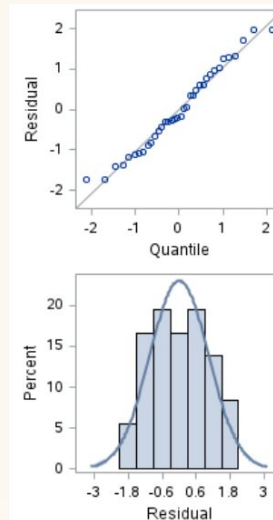
Influential Observations:
Cook's Distance values were all well below commonly used concern thresholds, indicating 2 potential influential points.



Independence: Residuals versus time slot showed no clear trend or clustering, supporting the assumption of independent observations across trials.



Constant Variance:
Residuals versus predicted values displayed relatively similar spread across fitted absorbency levels, suggesting no major heteroscedasticity.



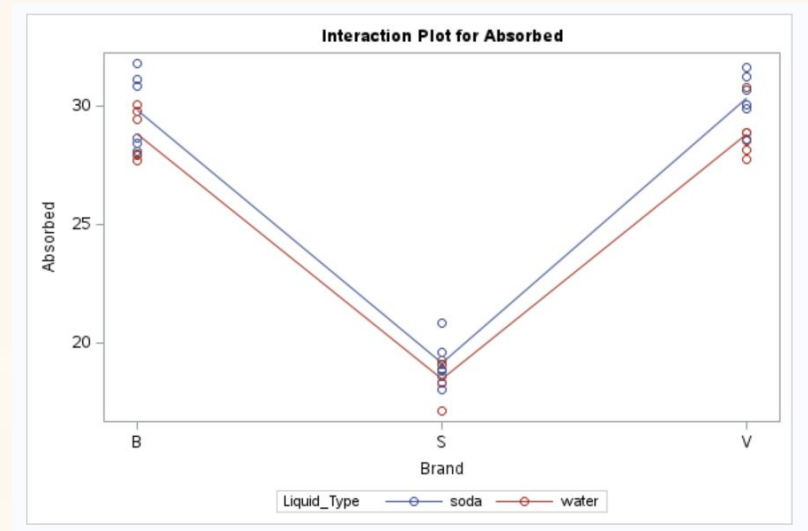
Normality: The Q-Q plot and residual histogram indicated approximate normality, with only minor deviations at the tails.

Main and interaction effects

Type III Tests of Main/interaction effects

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Brand	2	897.6552000	448.8276000	356.14	<.0001
Liquid_Type	1	9.8910250	9.8910250	7.85	0.0088
Brand*Liquid_Type	2	1.2098667	0.6049333	0.48	0.6234

- The interaction between brand and liquid type was not statistically significant, $F(2,30)=0.48$, $p=0.6234$, so there is not enough evidence that the effect of brand depends on liquid type. This is further supported by the interaction plot which shows parallel lines.
- The main effect of brand was significant, $F(2,30)=356.14$, $p<0.0001$, indicating that at least one brand mean differed from the others.



Block Assessment

- Liquid type was used as a blocking factor to control experimental variation. Since $MS(\text{Liquid_type}) = 9.89$ is much larger than $MSE = 1.26$, liquid type explained meaningful variation in absorbency. This supports that blocking was effective in reducing unexplained error.

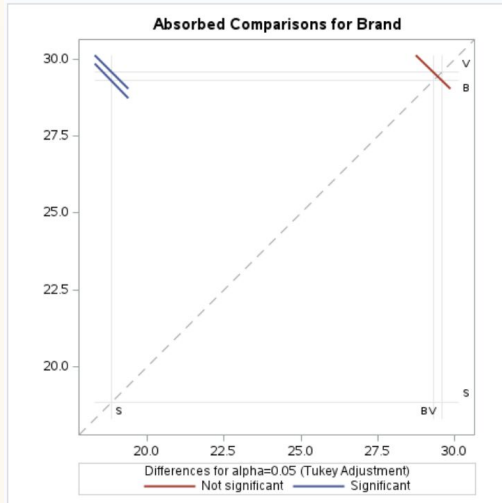
ANOVA results for GCBD

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	908.7560917	181.7512183	144.22	<.0001
Error	30	37.8073833	1.2602461		
Corrected Total	35	946.5634750			

Liquid_Type	1	9.8910250	9.8910250	7.85	0.0088
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How Do Towel Brands Differ in Absorbency?

Brand Codes: 1 = B, 2 = S, 3 = V



Brand	Absorbed LSMEAN	95% Confidence Limits	
B	29.295833	28.633997	29.957670
S	18.845833	18.183997	19.507670
V	29.575833	28.913997	30.237670

Least Squares Means for Effect Brand				
i	j	Difference Between Means	Simultaneous 95% Confidence Limits for LSMean(i)-LSMean(j)	
1	2	10.450000	9.320168	11.579832
1	3	-0.280000	-1.409832	0.849832
2	3	-10.730000	-11.859832	-9.600168

Interpretation:

- Brand S is statistically significantly lower than Brand B and V.
- Brand B and V were not significantly different from each other since their confidence interval includes 0.
- Highest Performance: Brands V (29.58g) and B (29.30g) achieved the highest similar mean absorbency values.

Results and Interpretation

Results

Brand	Price per roll	Price per sheet
 Viva	\$ 3.28	\$ 0.026
 Bounty	\$ 3.50	\$ 0.028
 Sparkle	\$ 1.69	\$ 0.015



Answers to the original research questions:

- After controlling for liquid type using a GCBD, towel brand has a significant effect on liquid absorption.
- Viva and Bounty absorb significantly more liquid than Sparkle.



The Best Performers:

Viva and Bounty are the most absorbent.



The Underperformer:

Sparkle is significantly less absorbent than the other two brands.



Practical implications:

- **Performance:** Viva and Bounty are statistically identical top-performers.
- **Value:** Sparkle is cheaper per roll, but its lower capacity may require using twice as many sheets.

Discussion and Conclusion

Summary:

- We have strong statistical evidence that paper towel brand significantly affects the amount absorbed ($p < 0.001$).
- **Viva** and **Bounty** are the most **effective**; Under tukey procedure, we do not have statistical evidence to conclude that they differ between both brands.
- **Sparkle** is significantly **less** absorbent, holding approximately 10.5g less liquid than the leading brands.
- With $MSE < MS(\text{liquid_type})$, we can justify the use of blocking for our experiment.
- Viva was slightly cheaper than Bounty (\$3.28 vs. \$3.50) and had a slightly higher LSMeans absorption value (29.6 vs. 29.3), although this difference was not statistically significant.

Experimental Design Recap:

Blocking:

A general complete block design was used, blocked by liquid type. Each paper towel brands were tested within each block. Blocking was effective in reducing the unexplained variability in the experiment.

Randomization:

Within each liquid-type block, paper towel brands were randomized using the `sample()` function in R, ensuring an unbiased assignment of brands to time slots.

Replication:

Each paper towel brand was tested multiple times within each block, with 6 sheets per brand per liquid type, to provide repeated measurements and improve the precision and reliability of the results. The equal sample size created a more stable model which contributes to better estimates and comparisons.

Population Generalization:

We believe these results could benefit the food service industry by helping restaurants make more informed decisions about which paper towel brand to purchase. This application is supported by our choice of water and soda as blocking liquids, since these are common liquids encountered in restaurant settings.

Limitations:

Pepsi was the only soda used, which may not reflect performance of soda types in general. Due to time we decided to take only 2 blocking factors, and did not explore other potential material that may be of practical interest.

Possible Improvements or Future Work:

Expand Liquid Variety: Test different liquid viscosities and sugar contents, such as oils, syrups, or milk, to see if the "identical top-performers" (Viva and Bounty) maintain their lead under tougher conditions.

Test under more realistic conditions: Test under more realistic conditions by changing the data collection method to measure the number of paper towel sheets needed for a worker to effectively clean a spill.

Group Contributions:

Julio: Experiment: Liquid administration on paper towel. **Slide presenter:** Sample Size Calculation + Results, Discussion and Conclusion

Xueying: Experiment: Towel weigher, trial set up. **Slide presenter:** Experimental Design + Data Collection

Stella: Experiment: Record / note taker. **Slide presenter:** Introduction + Statistical Analysis

***All members worked well together discussing ideas and meaningfully contributed to every aspect of the project**